

Control System Theory Block Diagram Reduction

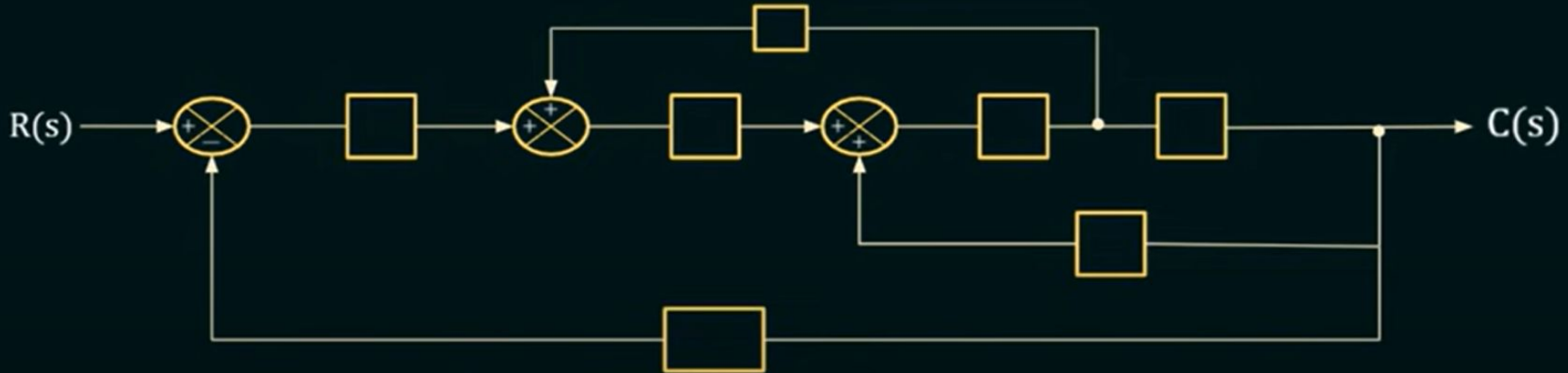
<https://youtube.com/playlist?list=PLBlnK6fEyqRineUwP-HPdkQvowXqq5IxA&si=1dXffwOVbx4EpPoL>

Block Diagrams

Block Diagrams are pictorial representation of a control system.

Block Diagram Representation is used to build a mathematical model of a control system which can be emulated on a computer.

Block diagram Representation is used to calculate the overall transfer function of the system.



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- ✎ Block Diagram Representation is used to build a mathematical model of a control system which can be emulated on a computer.
- ✎ Block diagram Representation is used to calculate the overall transfer function of the system.

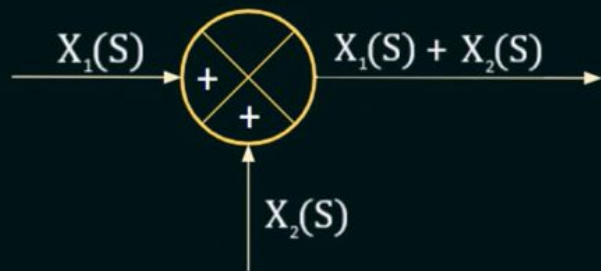
Elements of a Block Diagram:



- The signal into the block represents the input.
- The signal out of the block represents the output.
- The block itself represents the Transfer Function of the system.

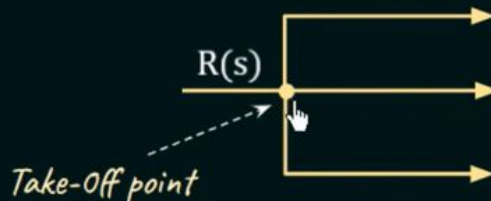
Summing Point:

A Summing Point/Summing Junction in a block diagram represents the dynamic summation of two (or more) signals.

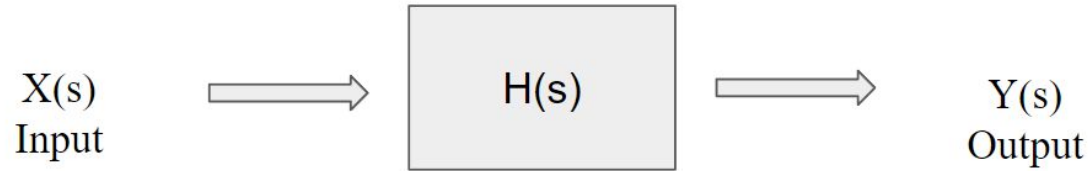


Take-off Point/ Pick-off Point/ Branch point:

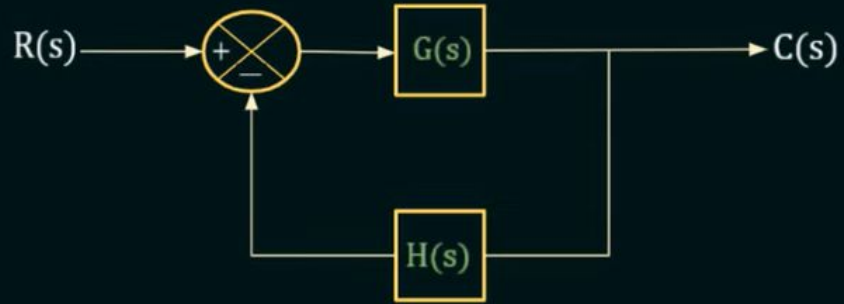
Take-off point in a block diagram represents a point where the signal branches out and goes concurrently to the other blocks or summing points.



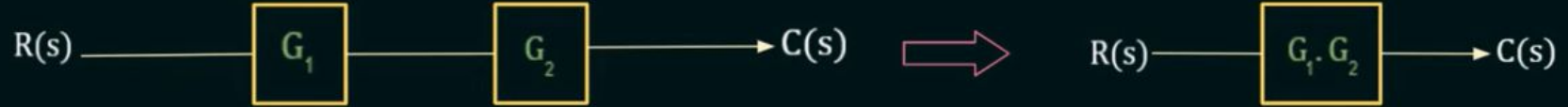
Block Diagram Reduction rule



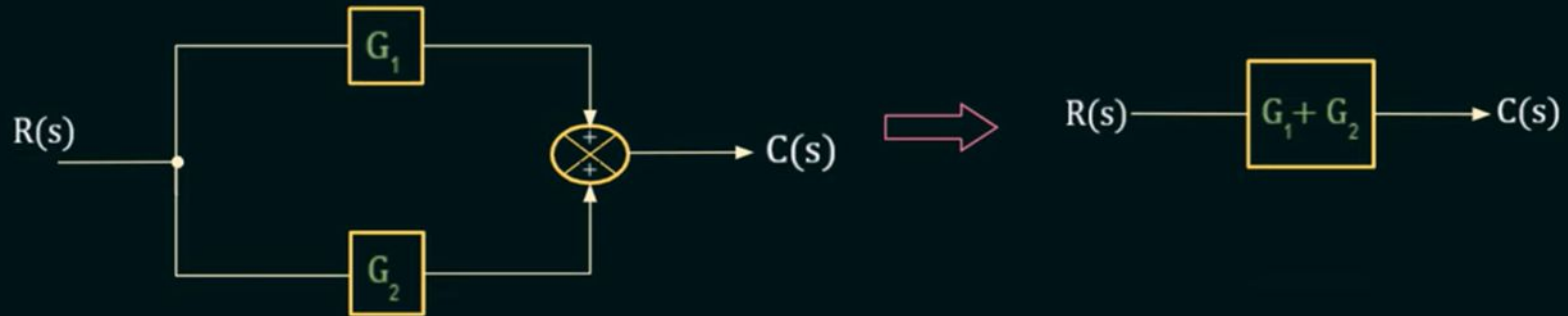
Rule 1: Representation of a closed loop system



Rule 2: When blocks are connected in series/cascade, their gains are multiplied.



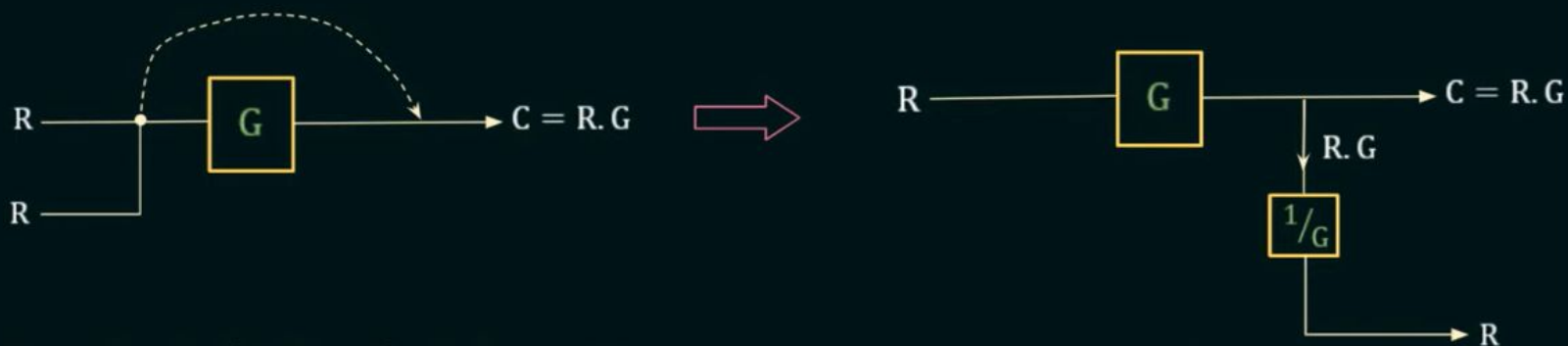
Rule 3: When blocks are connected in parallel, their individual gains are added.



Rule 4: Shifting of take-off point before a block.

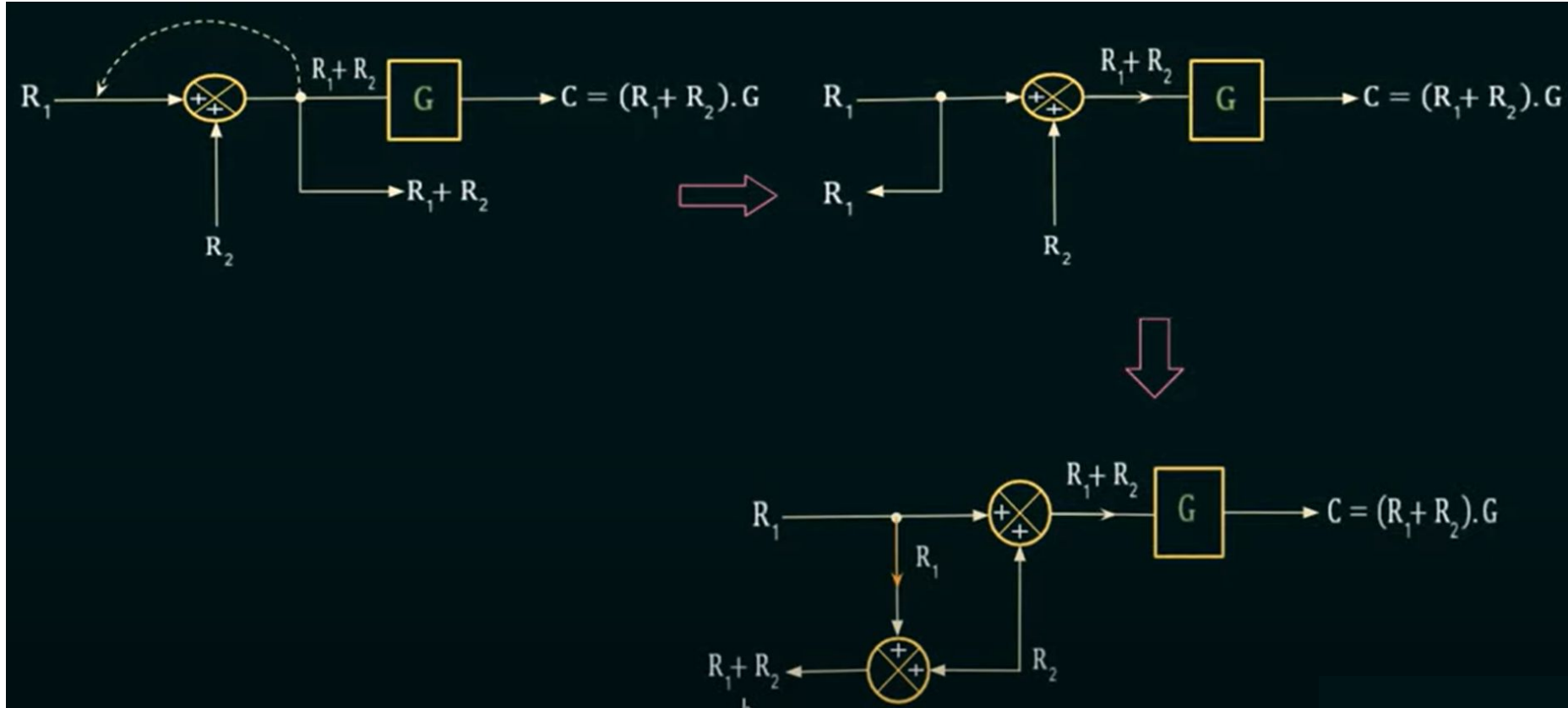


Rule 5: Shifting of take-off point after a block.

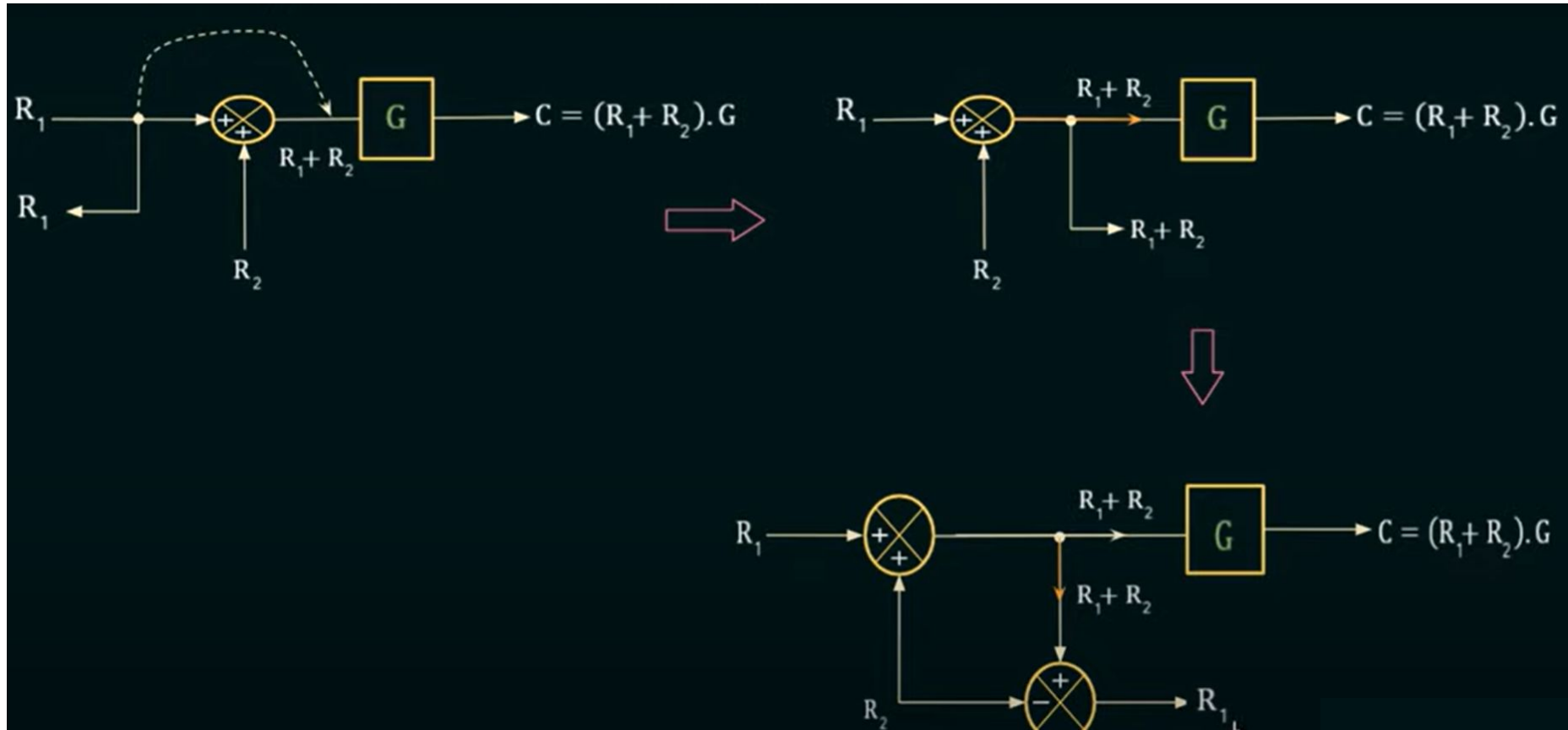


The output of take-off point must be same.

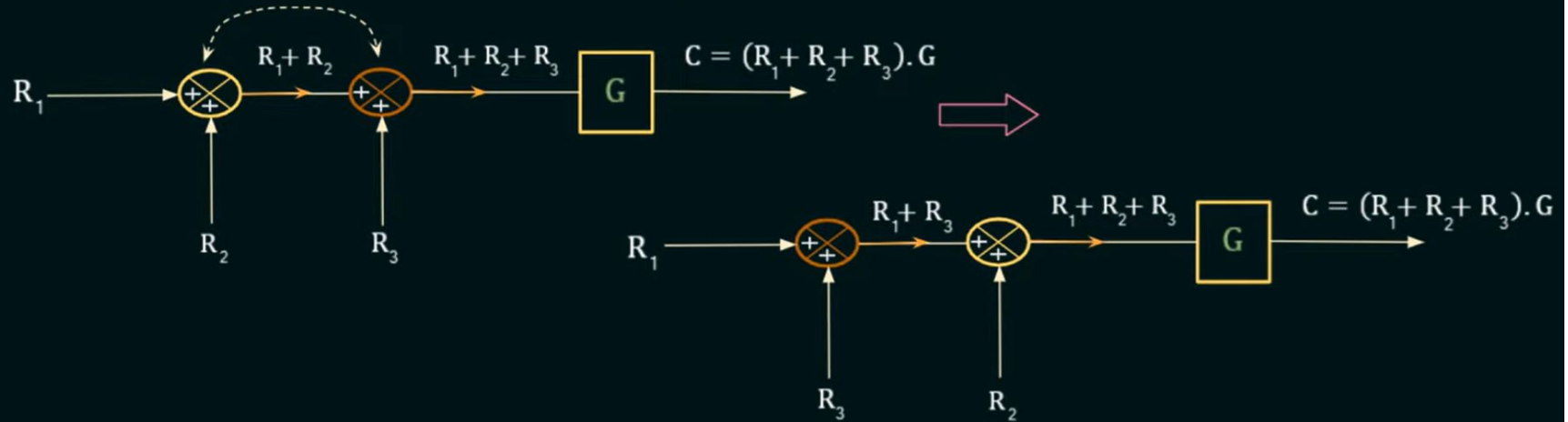
Rule 6 : Shifting take-off before an adder



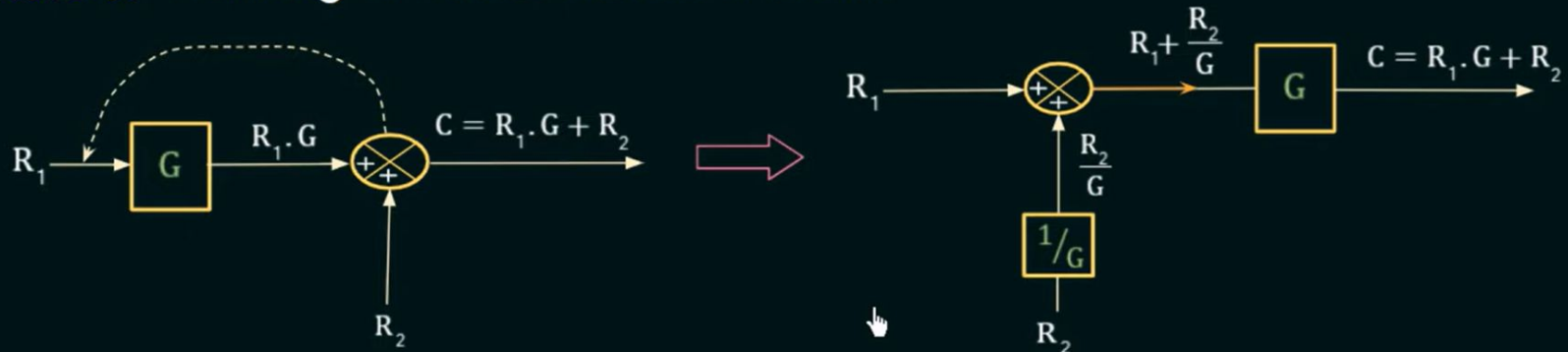
Rule 7: Shifting take-off after an adder



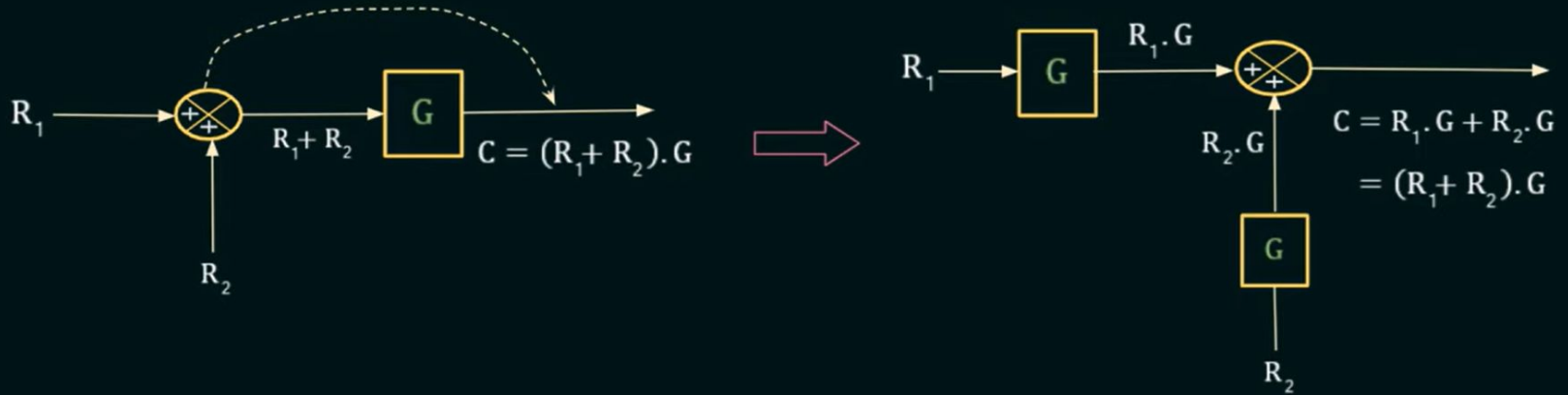
Rule 8: Rearrangement of adders.



Rule 9: Shifting of adder before a block.



Rule 10: Shifting of adder after a block.

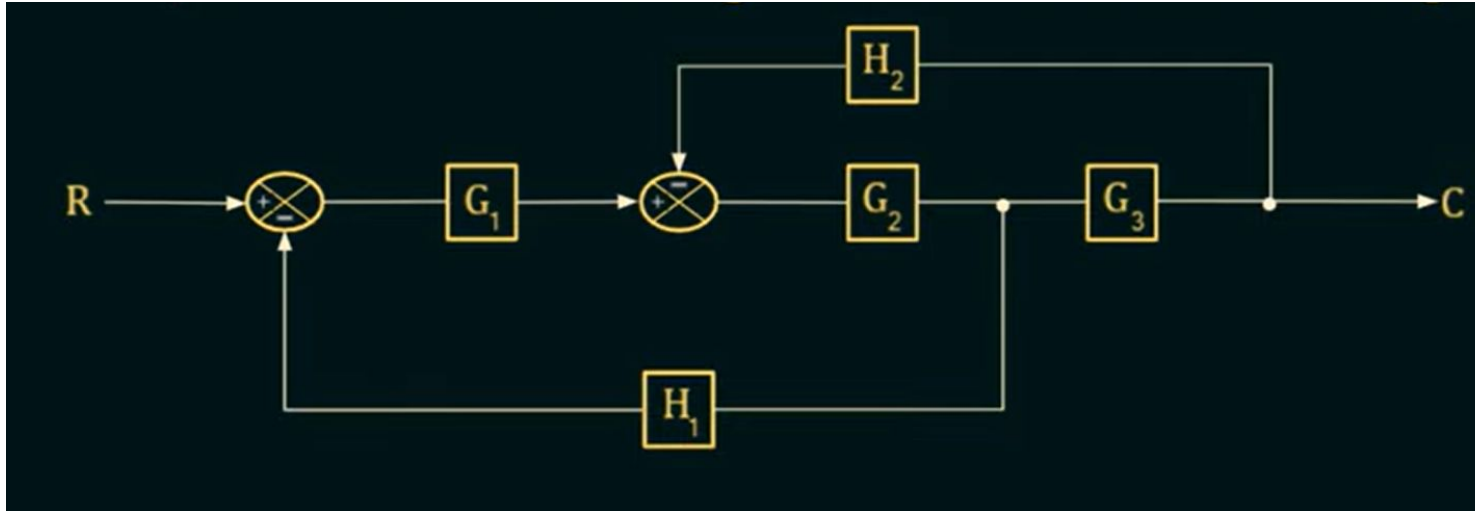


- If we shift the adder before a *block*, then we need to divide the gain of *block* from the input which is shifted.
- If we shift the adder after a *block*, then we need to multiply the gain of the *block* with the input which is shifted.

Block Diagram Reduction Algorithm

- **Rule 1** – Check for the blocks connected in series and simplify.
- **Rule 2** – Check for the blocks connected in parallel and simplify.
- **Rule 3** – Check for the blocks connected in feedback loop and simplify.
- **Rule 4** – If there is difficulty with take-off point while simplifying, shift it towards right/left.
- **Rule 5** – If there is difficulty with summing point while simplifying, shift it towards right/left.
- **Rule 6** – Repeat the above steps till you get the simplified form, i.e., single block.

Problem 1:



Rule 1 – Check for the blocks connected in series and simplify.

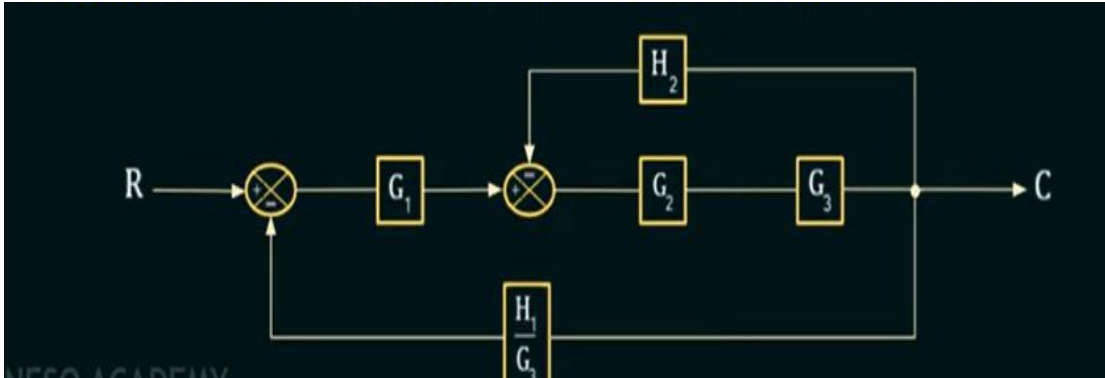
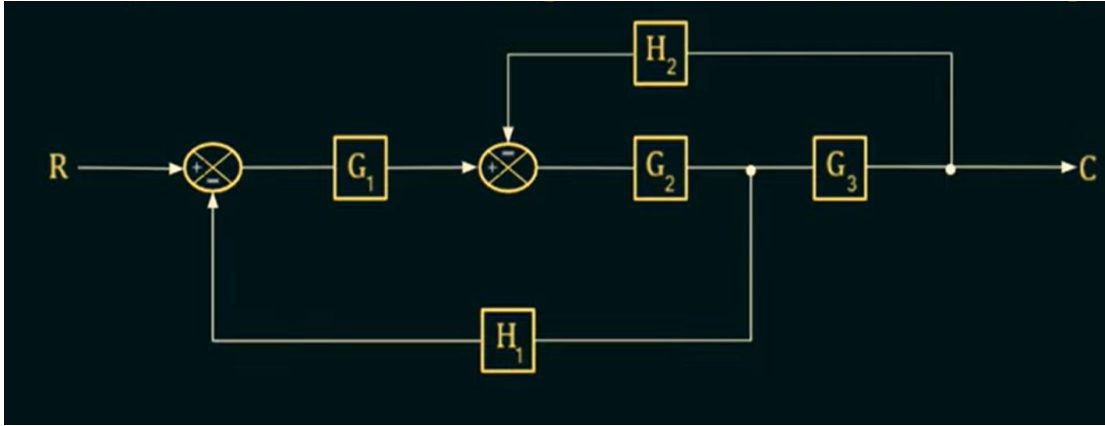
Rule 2 – Check for the blocks connected in parallel and simplify.

Rule 3 – Check for the blocks connected in feedback loop and simplify.

Rule 4 – If there is difficulty with take-off point while simplifying, shift it towards right/left.

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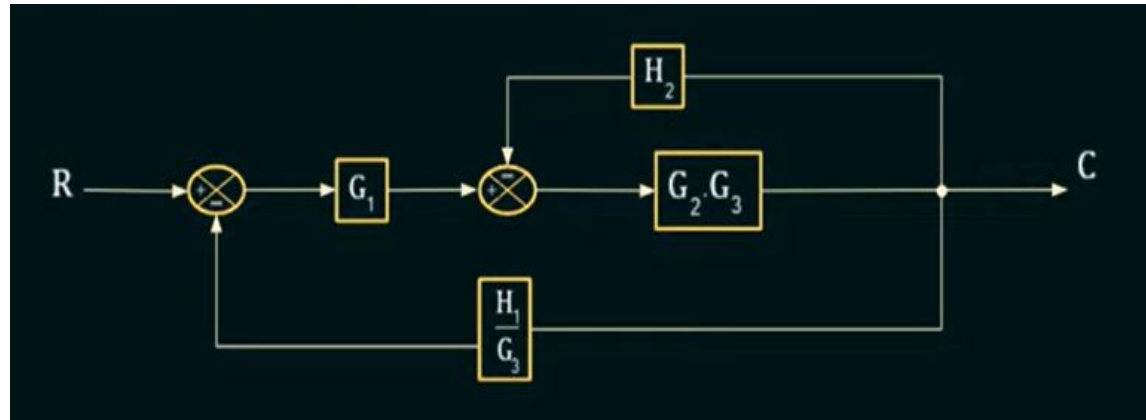
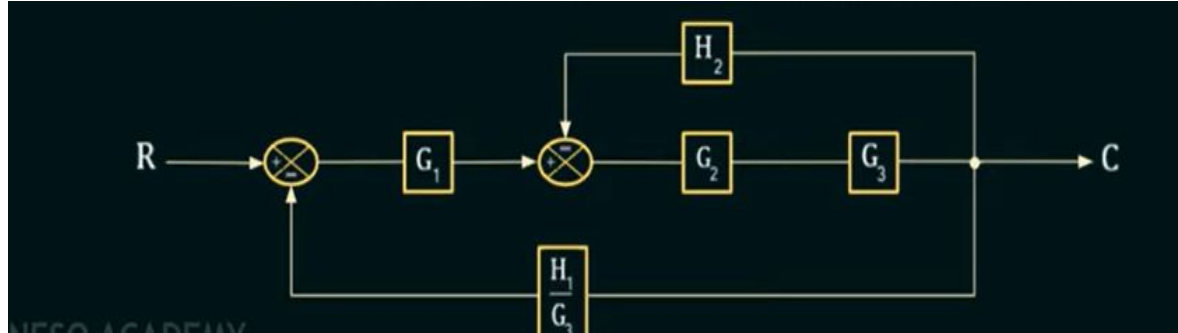
Rule 2 – Check for the blocks connected in parallel and simplify.

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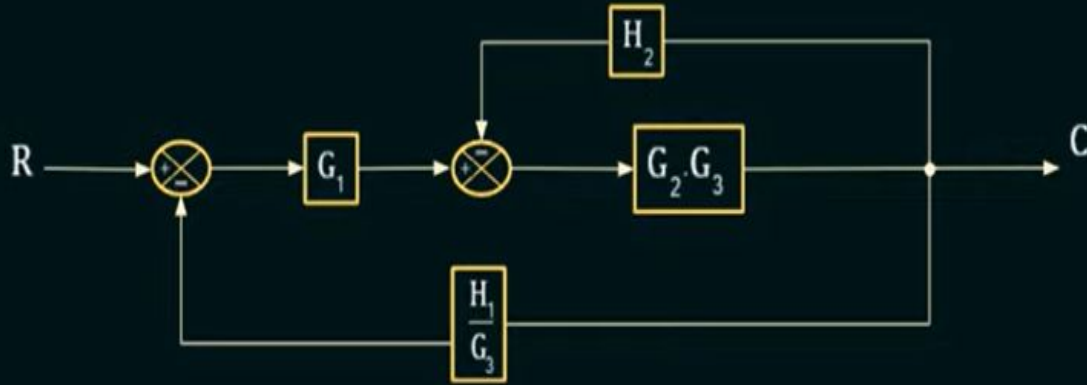
Rule 2 – Check for the blocks connected in parallel and simplify.

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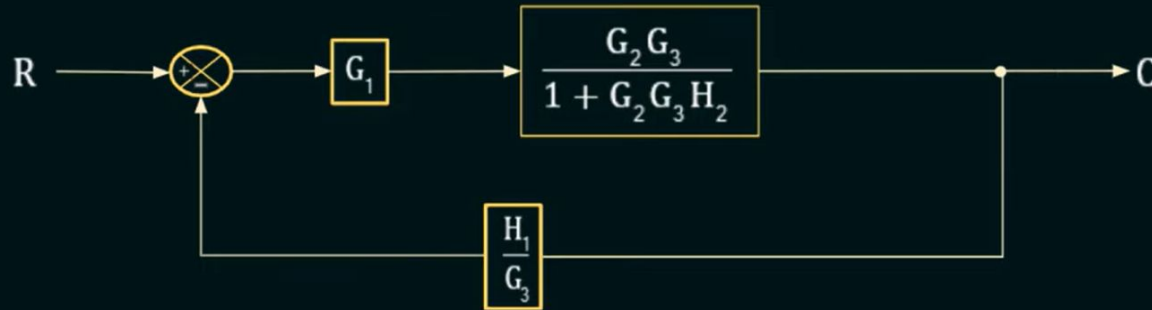
Rule 2 – Check for the blocks connected in parallel and simplify.

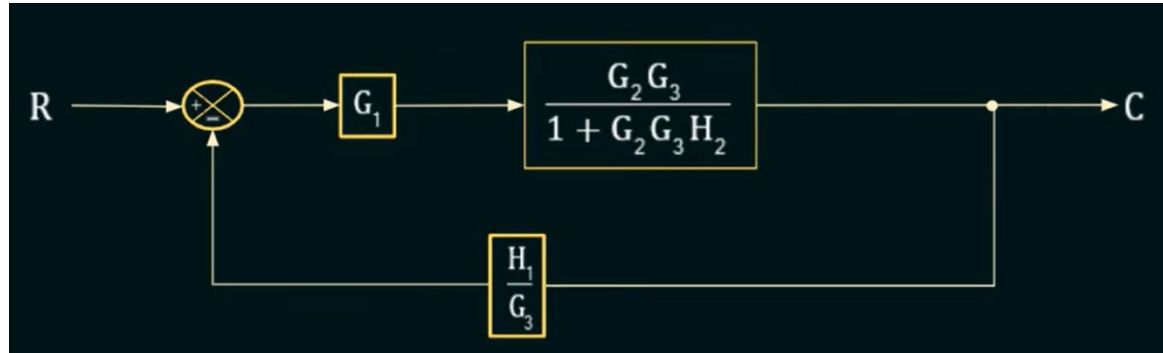
Rule 3 – Check for the blocks connected in feedback loop and simplify.

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Rule 1 – Check for the blocks connected in series and simplify.

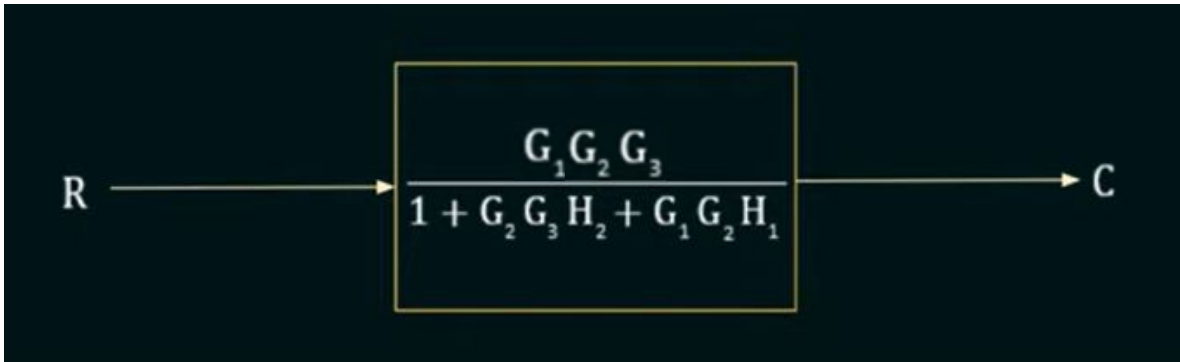
Rule 2 – Check for the blocks connected in parallel and simplify.

Rule 3 – Check for the blocks connected in feedback loop and simplify.

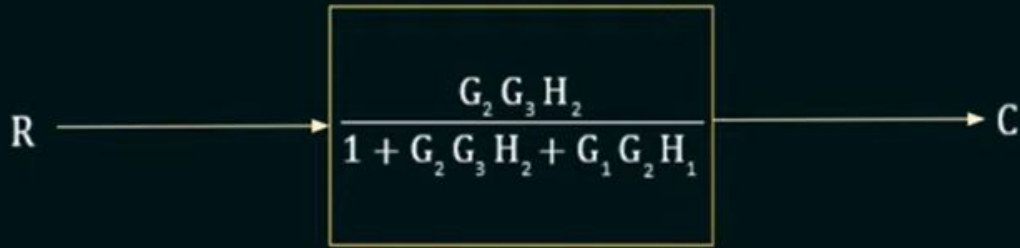
Rule 4 – If there is difficulty with take-off point while simplifying, shift it towards right.

Rule 5 – If there is difficulty with summing point while simplifying, shift it towards left.

Rule 6 – Repeat the above steps till you get the simplified form, i.e., single block.



Result



Rule 1 – Check for the blocks connected in series and simplify.

Rule 2 – Check for the blocks connected in parallel and simplify.

Rule 3 – Check for the blocks connected in feedback loop and simplify.

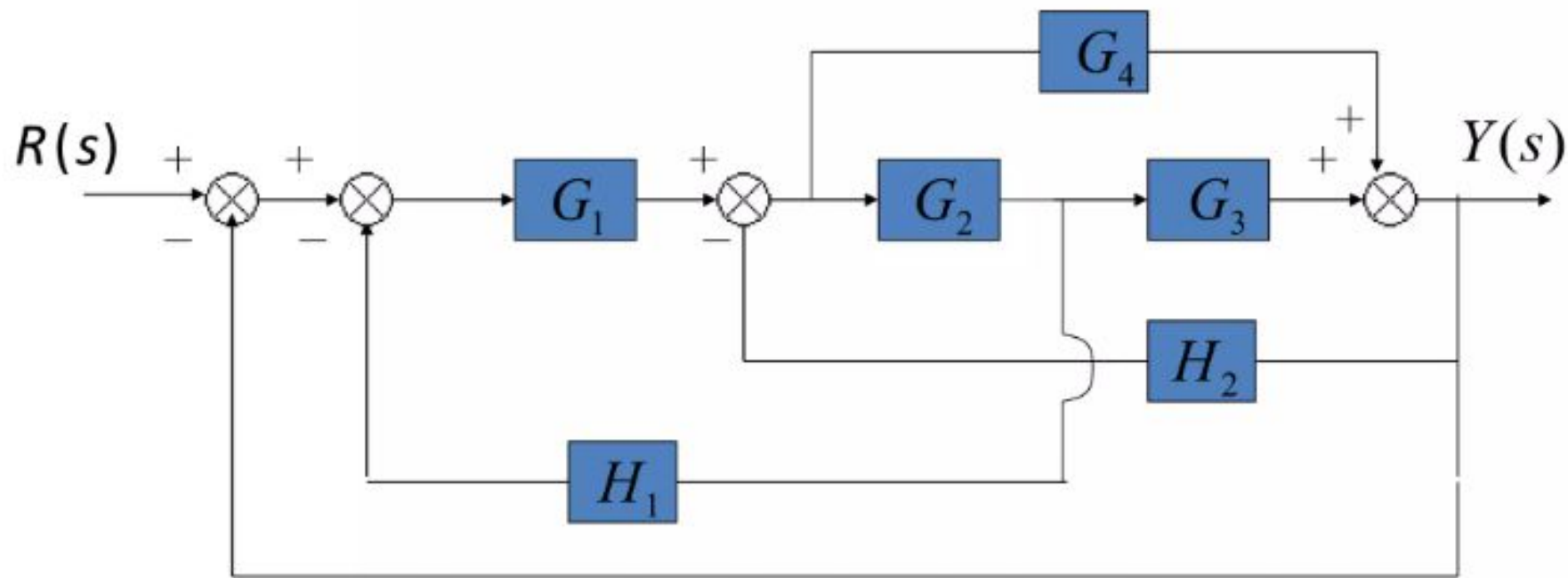
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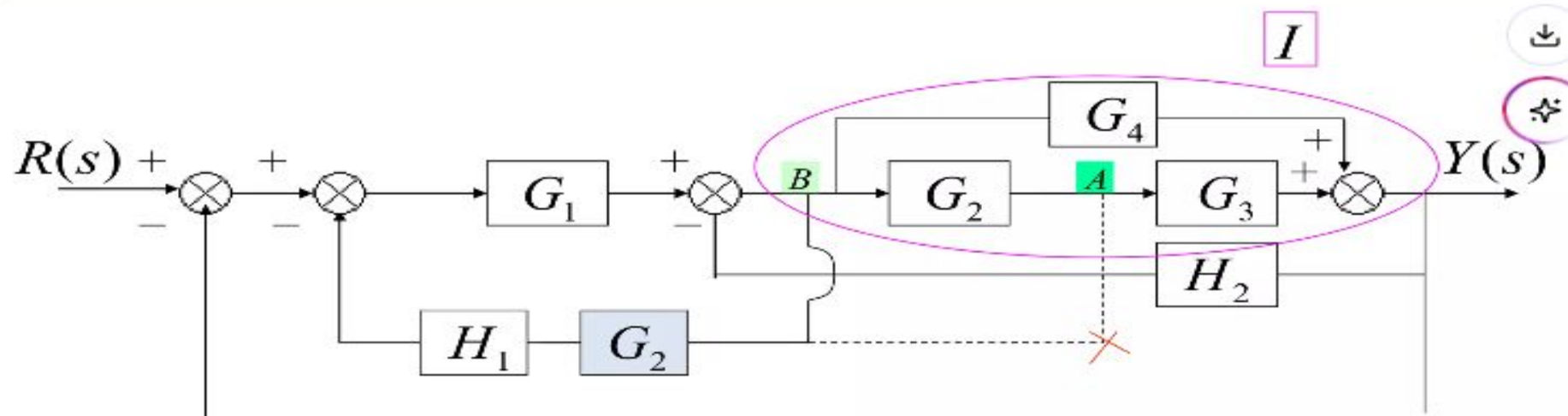
Rule 5 – If there is difficulty with summing point while simplifying, shift it towards left.

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Practice for Home

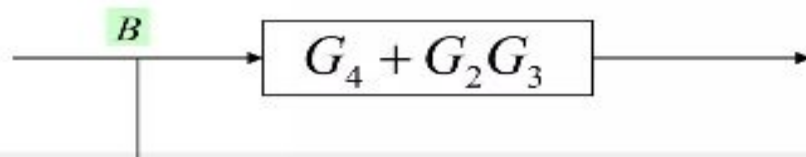
Problem 1

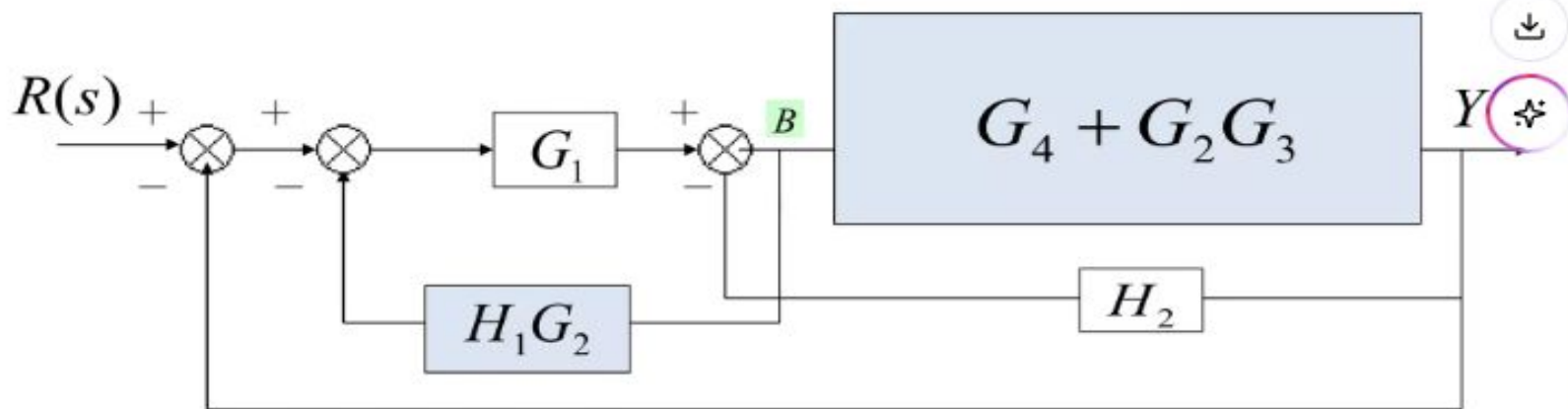




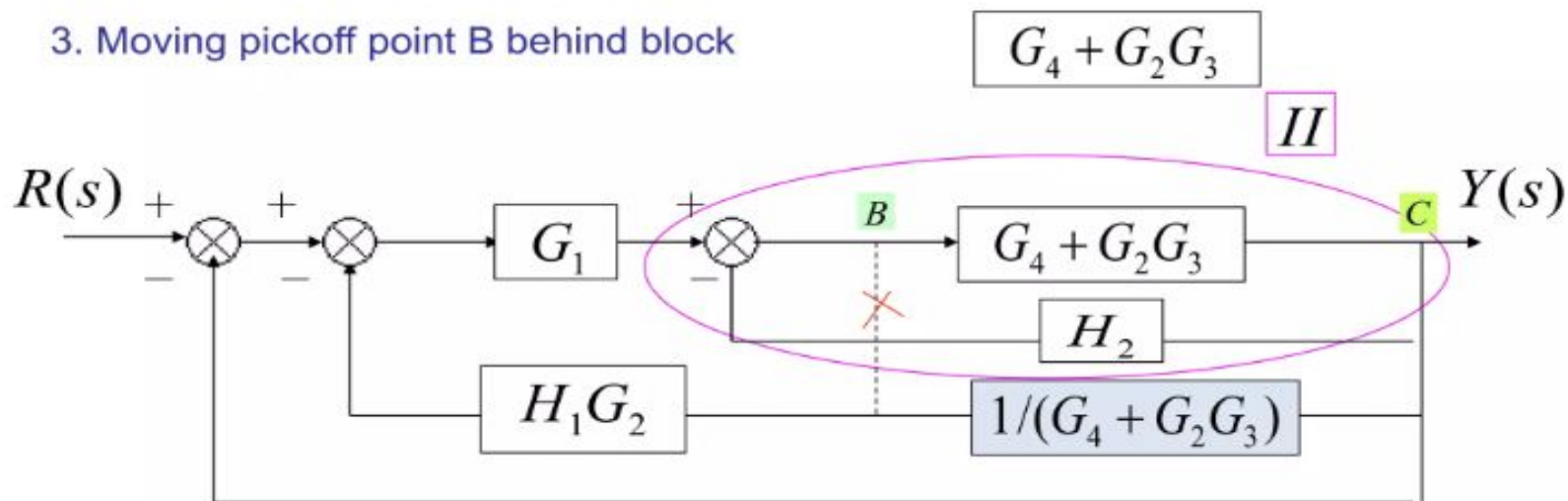
Solution:

1. Moving pickoff point A ahead of block G_2
2. Eliminate loop I & simplify

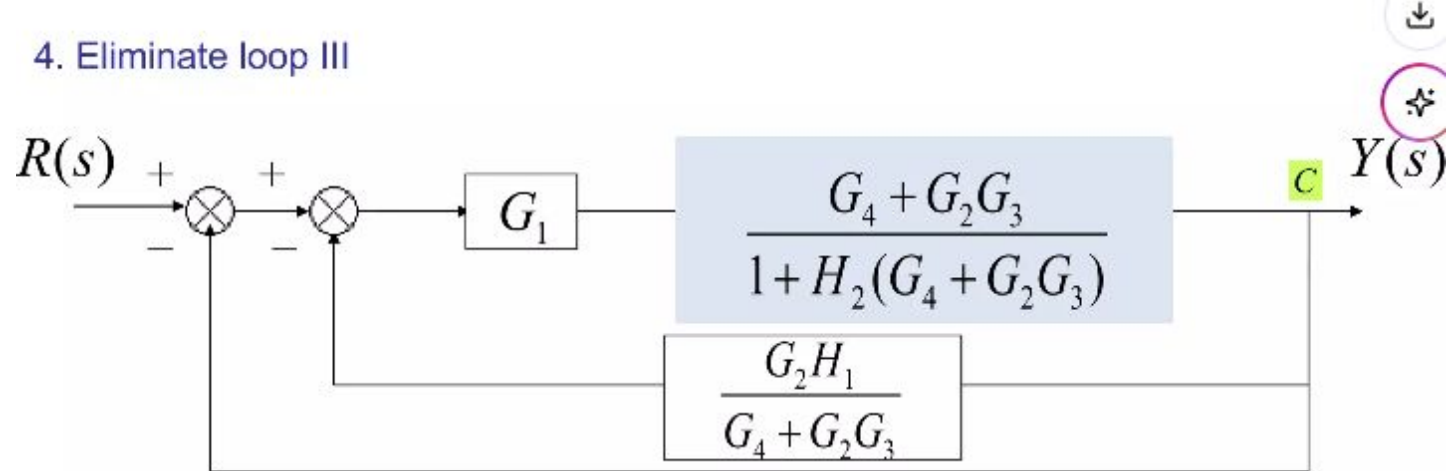




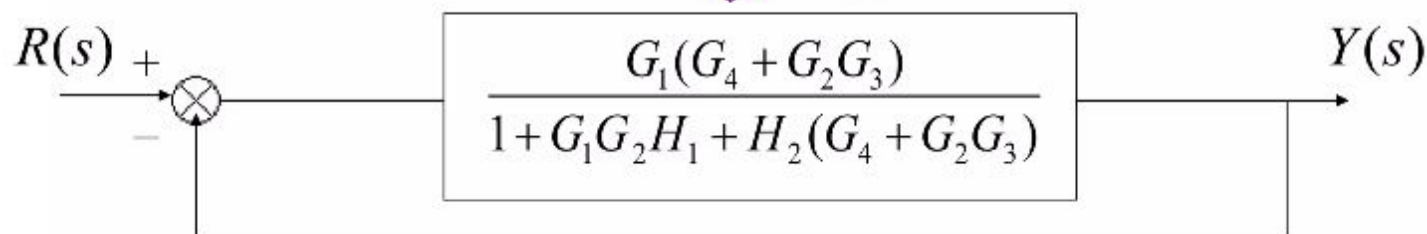
3. Moving pickoff point B behind block



4. Eliminate loop III



Using rule 6

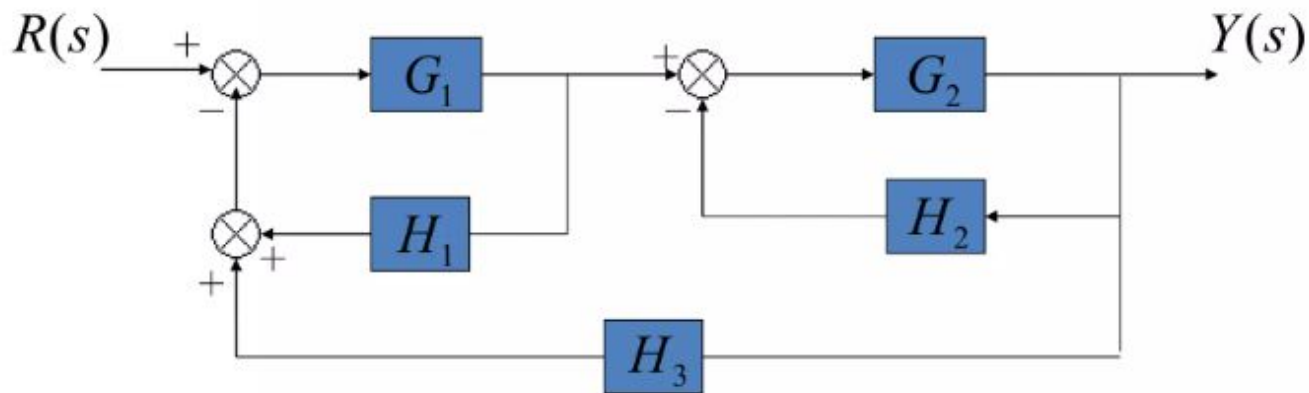


$$T(s) = \frac{Y(s)}{R(s)} = \frac{G_1(G_4 + G_2G_3)}{1 + G_1G_2H_1 + H_2(G_4 + G_2G_3) + G_1(G_4 + G_2G_3)}$$

Problem 2

Example 10

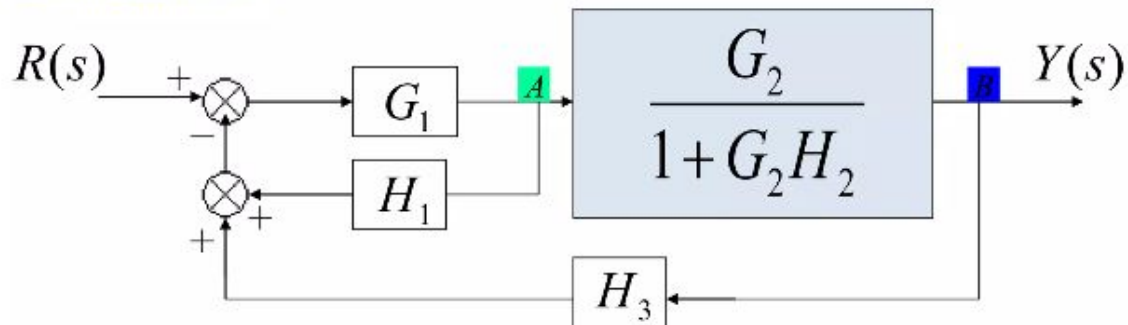
Find the transfer function of the following block diagrams



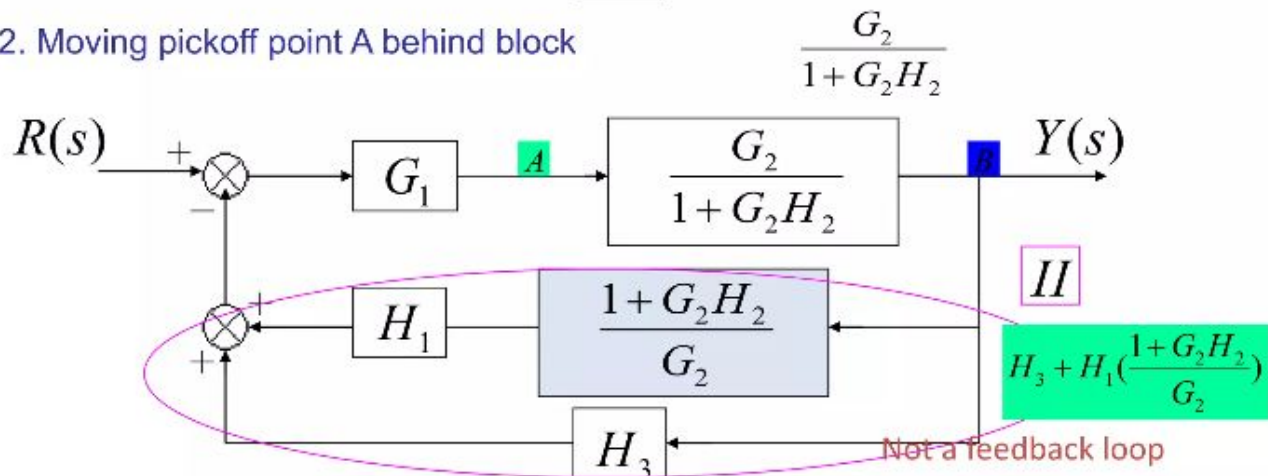
Solution:



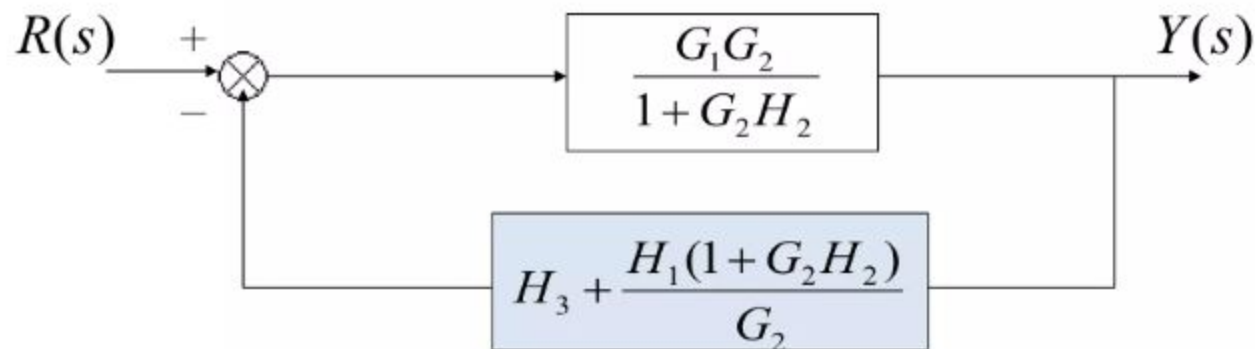
1. Eliminate loop I



2. Moving pickoff point A behind block



3. Eliminate loop II



Using rule 6

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G_1 G_2}{1 + G_2 H_2 + G_1 G_2 H_3 + G_1 H_1 + G_1 G_2 H_1 H_2}$$

Next Class

PID Control

Thank You